Stellar Cyber 6.6.0s is now live! Discover all the updates in the release notes!

Configuring Standard Sensor Profiles 6.5.0

Learn more at Stellar Cyber Academy.

The following links take you to courses on the Stellar Cyber Academy technical training portal where you can learn more about a topic by watching the suggested lessons.

M06 Open XDR Linux Server Sensor Installation and Configuration (01h:21m)

Enhanced FIM (18m:00s)

Learn how to enable and configure File Integrity Monitoring (FIM) on Linux Server Sensors within the Stellar Cyber Platform, setting up detection for file creation, deletion, and modification events to strengthen security monitoring. Follow along with a hands-on demonstration that walks through the configuration process and validates file monitoring accuracy, confirming that FIM events are captured correctly and surfaced as Interflow records in the data lake for analysis.

M05 Open XDR Windows Server Sensor Installation and Configuration (02h:45m)

Advanced Configuration - Sensor Profile (18m:00s)

Customize sensor profiles using predefined templates that align with specific security requirements, enabling granular control over data collection and processing. Watch a demo on selecting and applying templates, fine-tuning event settings, and monitoring enhanced data flow to the data processor within the Stellar Cyber Platform.

Advanced Configuration - File Integrity Monitoring (FIM) (18m:00s)

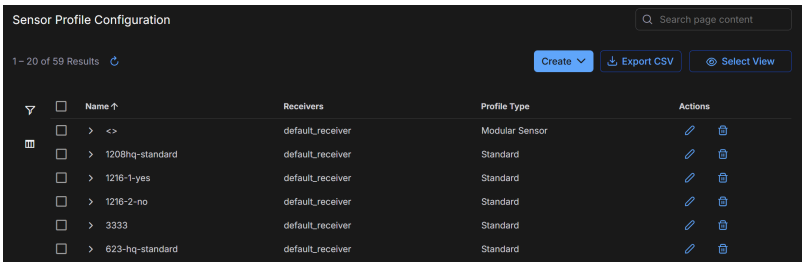
Enable File Integrity Monitoring (FIM) on the Stellar Cyber Platform to track and detect unauthorized modifications to critical files on Windows Servers. Learn to configure FIM settings for accurate file change tracking, then follow a hands-on demo to validate the setup and confirm that file change events are captured and forwarded correctly within the data lake for further analysis.

The first time you access a link on the portal during a session, you must log in to access content.

Whether you add or edit a standard sensor profile you have the same options, except you cannot edit the profile name. Standard sensor profiles can only be applied to standard sensors. To add a standard sensor profile:

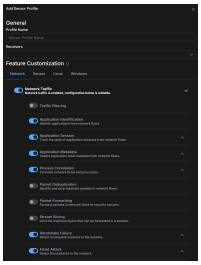
1. Select **System | DATA SOURCE MANAGEMENT | Sensors | Sensor Profiles**.

   The **Sensor Profile Configuration** page appears.



2. Select **Create** and choose **Add Standard Sensor Profile**.

   The **ADD SENSOR PROFILE** screen appears.



3. Enter the **Profile Name**.

   Stellar Cyber recommends that you establish a naming convention, so you can understand the intent of each profile by looking at the name. This field can only contain alphanumeric characters, underscores, spaces, and dashes.

4. Choose at least one **Receiver**.

   Each sensor profile must have at least one receiver, which is the destination of the data it collects.

   See the Receiver configuration page for more information on working with receivers.

5. Customize your profile settings in the available tabs. The available settings are explained below.

   a. **CPU Limit (%)** – Specify the amount of CPU per core that the sensor can use. To get the total for the sensor, multiply this percent by the number of cores. For example, if you set this at 5% and you have 8 cores, the sensor could use up to 40% of the CPU, as reported by the `top` and `htop` utility (distributed). If you have 32 cores, the sensor could use up to 160%. A total above 100% might seem impossible, but it's the sum of the usage percent on all the cores, each of which can have a maximum limit up to 100%.

   Note that the `aella_flow` process uses multiple threads. When using the `top` or htop utility with thread-mode enabled (`top H` or `htop H`), the output will show the percent that both the `aella_flow` process and its children threads use. At first, this might look like the CPU is processing twice as much as is configured because the individual threads add up to the same amount as their

parent process and they're all shown. To see the output displayed hierarchically so you can visually distinguish data for the parent process from data for its children threads, enable tree view mode (`top V` or `htop F5`).

The commands to use:

- In `top`, press **H** to show threads and **V** to enable tree view.
- In `htop`, press **H** to show threads and **F5** to enable tree view.

By enabling the thread display along with tree view, you can clearly see which threads are associated with which parent processes, providing a detailed and organized view of CPU usage.

> The Server Sensor uses a minimum of 300 MHz, even if the value specified for **CPU Limit (%)** results in a smaller value based on the host CPUs. For example, if the Server Sensor is installed in a host with 2 CPUs running at 2.2 GHz and you specify a value of **5%** for **CPU Limit (%)**, the arithmetic of **5% * 2 CPUs * 2.2 GHz** would indicate an assignment of **220 MHz** for the Server Sensor. However, in a situation such as this, the Server Sensor uses its minimum assignment of 300 MHz, even though it is greater than the specified value.

6. Use **Restore Windows Profile to Template** to apply an available template to your profile settings in the **Windows** tab.

   Separate templates let you apply **Low**, **Medium**, and **High volume** settings for Windows Server Sensors, depending on your deployment needs. Refer to **Windows Server Sensor Customizations** for details on the available settings.

7. Select **Submit**.

   The profile is immediately active.

Customizing Standard Sensor Profile Features

The available customizations are grouped into tabs under **Feature Customization**, providing control over:

- Network Tab – Network traffic analysis settings for device sensors and Linux Server Sensors
- Sensor Tab – Buffering settings
- Linux Tab – Settings for Linux Server Sensors
- Windows Tab – Settings specific to Windows Server Sensors

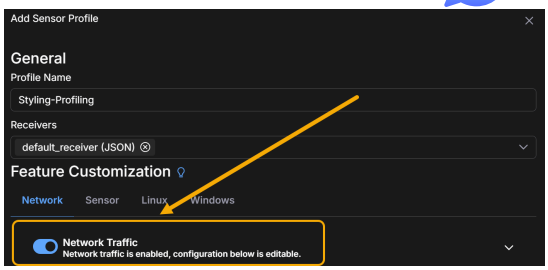
To enable or disable an option or group of options within the tab, use the toggle for that option. To open a group of options to fine-tune your customization, select the name of the group (or anywhere else away from the toggle).

Sensor Profile Customizations in the Network Tab

These settings apply to network sensors and Linux Server Sensors.

Network Traffic

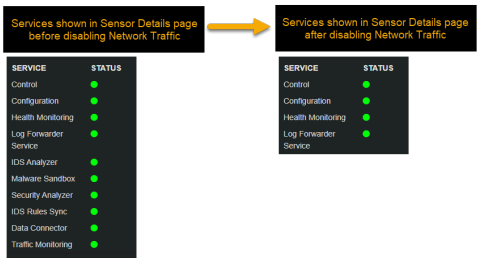
You can toggle the analysis of N ork Traffic on and off as a whole in the **Network** tab:



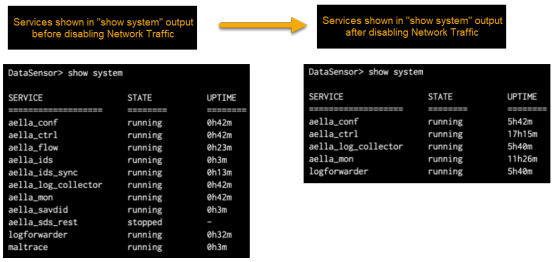
With **Network Traffic** disabled, the sensor operates as a Log Forwarder. None of the other features in the **Network** tab are available for selection or configuration. In addition:

- **Malware Sandbox** is disabled in the **Sensor** tab.
- **IDPS** is disabled in the **Sensor** tab.

You can see the differences in the system processes running on a sensor before and after disabling **Network Traffic** in a sensor's Sensor Details page, as shown in the illustration below:



You can see similar information with the `show system` command in the sensor CLI. For example, the `show system` processes at left are for a sensor with all main sensor features enabled (**Network Traffic**, **Malware Sandbox**, **IDPS**, and **Log Forwarder**). By contrast, the `show system` output at right shows a sensor after **Network Traffic** has been toggled off:



When you disable Network Traffic, Stellar Cyber ask you to confirm your decision and provides you with the option of saving or clearing buffered data for affected sensors.

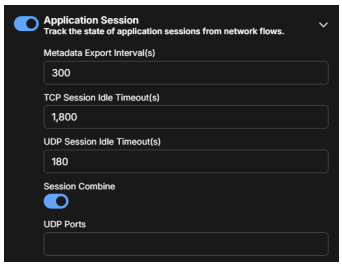
Traffic Filtering (Pre-DPI Filters) 5.5.0

Enable this option to optimize sensor performance by removing unnecessary traffic before performing Deep Packet Inspection (DPI). For configuration details, see Pre-DPI Filters .

Application Identification

Enable this option for the sensor to identify applications associated with events.

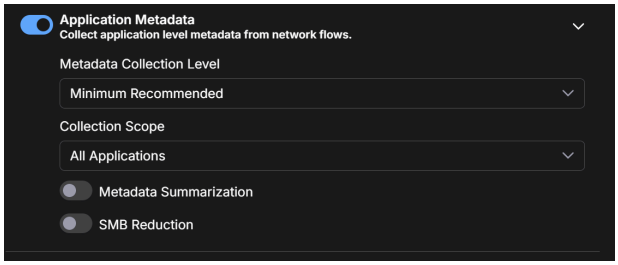
Application Session



Enable this option for the sensor to identify the length of individual network sessions. The parameters specify how often to report the session information and what timeout to apply when determining that a session has ended.

- If you disable **Application Session**, network and server sensors stop sending data to the DP. Data sensors continue to process traffic.
- If you enable **Session Combine**, UDP sessions from the specified ports to the same destination are combined into a single session.

Application Metadata



Enable this to control how application information is collected and which applications are included.

You can disable the collection of application metadata to increase sensor performance and save storage space, but that limits collected data to:

- Basic session information
- Packet and byte counts
- IP address
- Port

You can set the **Metadata Collection Level** to **Limited Visibility**, **Minimum Recommended**, **Standard**, or **Maximum**. This controls how many of the information fields are collected.

Click to see the fields collected by **Limited Visibility**:

- 0zz0 login
- 0zz0 password
- activesync login
- aim_express contact_login
- aim_express login
- aims login
- archive login
- archive password
- badongo login
- badongo password

- badoo contact_login
- badoo login
- badoo password
- bebo login
- bebo password
- dhcp agent
- dhcp chaddr
- dhcp ciaddr
- dhcp client_fqdn
- dhcp domain_name
- dhcp dns_server
- dhcp end
- dhcp end_status
- dhcp giaddr
- dhcp host_name
- dhcp inherit_parent
- dhcp ip_lease_time
- dhcp message_type
- dhcp option
- dhcp option_type
- dhcp option_value_buffer
- dhcp request
- dhcp router
- dhcp siaddr
- dhcp sname
- dhcp subnetmask
- dhcp xid
- dhcp yiaddr
- dimp login
- dimp password
- dns dns_entry
- dns dns_query
- dns end
- dns tunneling
- dns flags
- dns host
- dns host_addr
- dns host_addr6
- dns host_type
- dns message_type
- dns name
- dns query
- dns query_type
- dns reply_code
- dns reverse_addr
- dns reverse_addr6

- dns section_type
- dns transaction_id
- dns ttl
- ebuddy contact_login
- ebuddy login
- ebuddy password
- facebook login
- facebook password
- facebook_mail login
- friendster login
- friendster password
- ftp content_type
- ftp data_port
- ftp filename
- ftp filesize
- ftp index
- ftp loadway
- ftp login
- ftp password
- ftp method
- ftp return_code
- ftp transfer_duration
- gmail_basic login
- gmail_chat login
- gmail_mobile login
- gmail_mobile password
- gmx login
- hi5 login
- hi5 password
- http auth_username
- http auth_password
- http code
- http index
- http method
- http proxy_auth
- http proxy_login
- http proxy_realm
- http referer
- http server
- http uri
- http filename
- http file_type
- http forward_addr
- http forward_addr6
- http host
- http mime_type

- http part_content_type
- http user_agent
- ica login_info
- ica login_info_utf16
- icmp code
- icmp message
- icmp rtt
- icmp type
- kakaotalk login
- krb5 error_code
- krb5 login
- krb5 server
- krb5 message_type
- krb5 service
- laposte_webmail login
- ldap hostname
- ldap name
- linkedin login
- live_groups login
- live_hotmail login
- livemail_mobile login
- mailru login
- maktoob login
- mimp login
- mimp password
- msn contact_login
- msn login
- mysql login
- mysql query
- mysql request
- mysql result_set
- mysql sqlstate_code
- mysql number_columns
- mysql number_rows
- mysql query_id
- netflix login
- netlog login
- netlog password
- nfs filename
- nfs filesize
- nfs type_string
- nfs symlink_name
- nfs uid
- niconico_douga login
- nntp login
- nntp password

- odnoklassniki login
- odnoklassniki passwd
- oovoo login
- orangemail login
- owa login
- paltalk contact_login
- paltalk login
- paltalk_transfer login
- pap login
- pap password
- perfspot login
- perfspot password
- pop3 password
- postgres authentification_type
- postgres error
- postgres login
- postgres password
- postgres query
- postgres query_id
- postgres sqlstate_code
- postgres number_columns
- postgres number_rows
- qq contact_login
- qq login
- qq_transfer login
- radius login_ipv6_host
- radius login
- rambler login
- rambler password
- rambler_webmail rambler_webmail
- rapidshare login
- rapidshare password
- rdp default_username
- rdp username_ascii
- rdp username_raw
- rlogin client_login
- rlogin server_password
- rsh login
- rsh remote_login
- rsync login
- rsync password
- rtmp page_url
- secondlife login
- skyblog login
- skyblog password
- slsk login

- slsk password
- smb login
- smb user_id
- smb nt_status
- smb command_string
- smpp receiver
- smpp sender
- smtp login
- smtp password
- smtp response_code
- snpp login
- snpp password
- socks5 login
- socks5 password
- spdy server_agent
- spdy user_agent
- squirrelmail login
- squirrelmail password
- ssh version
- ssl common_name
- ssl issuer
- ssl subject_alt_name
- ssl validity_not_after
- ssl version
- ssl validity_not_before
- tagged login
- tagged password
- tango login
- tchatche login
- tchatche password
- tds login
- tds password
- tds sqlstate_code
- tds query
- tds query_id
- tds number_columns
- tds number_rows
- teamspeak login
- teamspeak password
- telnet login
- telnet password
- tftp filename
- tftp filesize
- tftp query
- tftp request_filename
- tns login

- tns password
- tns query
- tns sqlstate_code
- twitter login
- twoo login
- twoo password
- ustream login
- ustream password
- viadeo login
- viadeo password
- vkontakte login
- vkontakte password
- winmx login
- yahoo_groups login
- yandex login
- yandex password
- yandex_webmail login
- ymail2 login
- ymail_classic login
- ymail_mobile_new login
- ymail_mobile_new password
- ymsg_conf login
- ymsg contact_login
- ymsg login
- ymsg_transfer login
- ymsg_video login
- ymsg_webmessenger contact_login
- ymsg_webmessenger login
- youtube login
- youtube password
- zimbra login
- zimbra password
- zimbra_standard login
- zimbra_standard password

Minimum Recommended collects everything in **Limited Visibility**. There is currently no difference between **Minimum Recommended** and **Limited Visibility**.

Standard collects everything in **Minimum Recommended** and:

- amqp class
- amqp type
- amqp method
- amqp replyto
- amqp routing_key
- dhcp circuit_id
- dhcp remote_id
- dhcp remote_id_subtype

- dhcp remote_id_type
- dns web_application_info
- dns ancount
- dns arcount
- dns nscount
- dns opcode
- dns qdcount
- ftp return_content
- ftp offset
- ftp inherit_parent
- ftp data_port_end_offset
- ftp data_port_start_offset
- ftp_data inherit_key
- http cache_control
- http connection_header
- http content_disposition
- http content_encoding
- http content_len
- http cookie
- http date
- http end
- http etag
- http expires
- http image_type
- http is_webdav
- http location
- http ntlm_domain
- http ntlm_user
- http ntlm_workstation
- http part_filename
- http part_header_name
- http part_header_value
- http part_name
- http request
- http request_size
- http server_agent
- http set_cookie
- http smb_client
- http version
- http video_type
- http video_url
- icmp seq
- icmp typeval
- icmp wrong_crc
- krb5 err_cname_name
- krb5 err_sname_name

- krb5 realm
- ldap krb5_enc_data
- ldap krb5_enc_data_type
- ldap krb5_err_cname
- ldap krb5_err_cname_name
- ldap krb5_err_cname_type
- ldap krb5_err_crealm
- ldap krb5_err_data
- ldap krb5_error_code
- ldap krb5_err_realm
- ldap krb5_err_sname
- ldap krb5_err_sname_name
- ldap krb5_err_sname_type
- ldap krb5_err_text
- ldap krb5_kdcoptions
- ldap krb5_message
- ldap krb5_message_type
- ldap krb5_pa_data
- ldap krb5_pa_data_type
- ldap krb5_pa_data_value
- ldap krb5_realm
- ldap krb5_server
- ldap krb5_service
- ldap krb5_ticket
- ldap krb5_ticket_enc_part
- ldap krb5_ticket_name
- ldap krb5_ticket_name_type
- mysql end
- mysql bind_variable
- mysql variable_id
- mysql variable_type
- mysql variable_value
- nfs current_state
- nfs gid
- nfs mode
- nfs procedure
- nfs status
- nfs type
- nfs version
- pop3 attach_filename
- pop3 file_type
- pop3 login_server
- pop3 login
- pop3 received_by_name
- pop3 received_date
- pop3 receiver

- pop3 received_from_name
- pop3 receiver_email
- pop3 sender
- pop3 sender_email
- pop3 subject
- postgres variable_format
- postgres variable_id
- postgres variable_type
- postgres variable_value
- postgres base
- postgres proto_version
- postgres server_version
- rpc message_type
- rpc procedure
- rpc program
- rpc xid
- sip authorization_username
- smb filename
- smb filesize
- smb file_type
- smb directory
- smb domain
- smb file_id
- smb host
- smb krb5_error_code
- smb krb5_err_text
- smb krb5_ticket
- smb loadway
- smb native_os
- smb path
- smtp attach_filename
- smtp attach_size
- smtp file_type
- smtp received_by_name
- smtp received_date
- smtp received_from_name
- smtp receiver
- smtp receiver_email
- smtp sender
- smtp sender_email
- smtp server
- smtp subject
- snmp method
- snmp name
- snmp oid
- snmp value_raw

- snmp version
- spdy location
- spdy mime_type
- ssl certif_sha1
- ssl handshake_type
- ssl organization_name
- ssl parent_common_name
- ssl protocol_version
- ssl server_name
- syslog code
- tds bind_variable
- tds variable_id
- tds variable_type
- tds variable_value
- telnet term_type
- tftp mode
- tns client_hostname
- tns server_hostname
- tns bind_variable
- tns variable_id
- tns variable_type
- tns variable_value

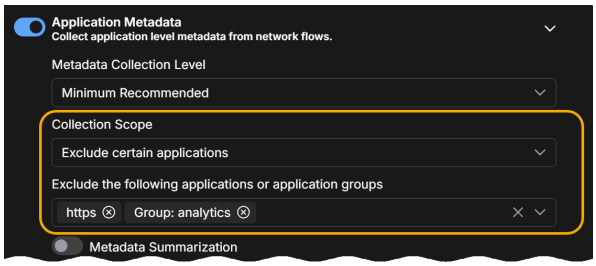
Maximum collects everything in **Standard** and:

- amqp correlation_id
- amqp exchange_type
- amqp revision
- amqp server_major_version
- amqp server_minor_version
- ica application
- ica application_utf16
- ica priority
- ica service
- mysql base
- pop3 attach_disposition
- pop3 attach_transfer_encoding
- pop3 date
- pop3 received_by_ip
- pop3 received_from_ip
- pop3 received_server_agent
- pop3 received_with
- pop3 receiver_alias
- pop3 receiver_type
- pop3 sender_alias
- pop3 user_agent
- rpc program_version
- rpc state

- rpc version
- sip inherit_key
- sip inherit_parent
- sip method
- sip reply_code
- sip request_call_id
- sip user_agent
- smb attributes
- smb ext_attributes
- smb file_attributes
- smb file_chunk_data_offset
- smb file_chunk_len
- smb krb5_enc_data
- smb krb5_enc_data_type
- smb krb5_err_cname
- smb krb5_err_cname_name
- smb krb5_err_cname_type
- smb krb5_err_crealm
- smb krb5_err_data
- smb krb5_err_realm
- smb krb5_err_sname
- smb krb5_err_sname_name
- smb krb5_err_sname_type
- smb krb5_kdcoptions
- smb krb5_message
- smb krb5_message_type
- smb krb5_pa_data
- smb krb5_pa_data_type
- smb krb5_pa_data_value
- smb krb5_realm
- smb krb5_server
- smb krb5_service
- smb krb5_ticket_enc_part
- smb krb5_ticket_name
- smb krb5_ticket_name_type
- smb search_attributes
- smb search_pattern
- smb security_blob
- smb service
- smb service_data
- smb session_key
- smb share_access
- smtp attach_disposition
- smtp attach_transfer_encoding
- smtp content_language
- smtp content_transfer_encoding

- smtp date
- smtp received_by_ip
- smtp received_from_ip
- smtp received_server_agent
- smtp received_with
- smtp receiver_alias
- smtp receiver_type
- smtp replyto
- smtp sender_alias
- smtp server_agent
- smtp start_time
- smtp stop_time
- smtp user_agent
- tns base
- tns client_os
- tns client_program_name
- tns client_program_path
- tns response_size
- tns server_os
- tns server_port
- tns version

Collection Scope allows you to filter specific metadata that is sent to the Data Processor.



- Select **All applications** for no traffic filtering
- Select **Exclude certain applications** to collect metadata for all content except the applications and application groups you specify in the selector box. .
- Select **Only collect certain applications** to exclude all metadata except that for the applications and application groups you specify in the selector box.

When you select an option other than **All applications**, the user interface updates for you to then **Exclude** or **Include the following applications or application groups.** Begin typing in the selector box to display a list of configured **Application Groups** and **Applications**. **Applications Groups** are listed first. Scroll through the list to get to the set of **Applications**. When you select an Application Group, the label for it is has a prefix (**Group:**) to distinguish it from Applications. In the image above, there is one application and one group.

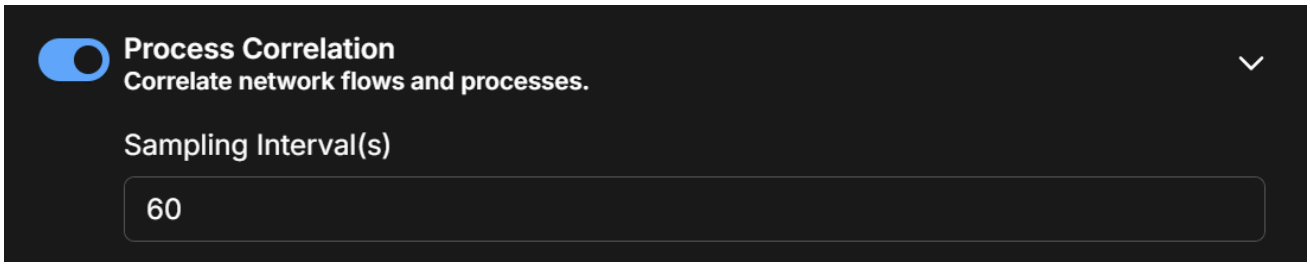
In previous releases, Application Groups were referred to as Tags.

Metadata Summarization allows you to reduce traffic to the DP. If you enable this, the sensor groups similar metadata from chatty applications into a single JSON file before sending to the DP.

Finally, you can enable **SMB Reduction** to reduce the amount of metadata collected for SMB commands. This is especially helpful in increasing your compression ratio if your network has a great deal of SMB traffic. If you enable this, metadata is only collected for the following:

- session setup
- logoff
- read
- write

Process Correlation

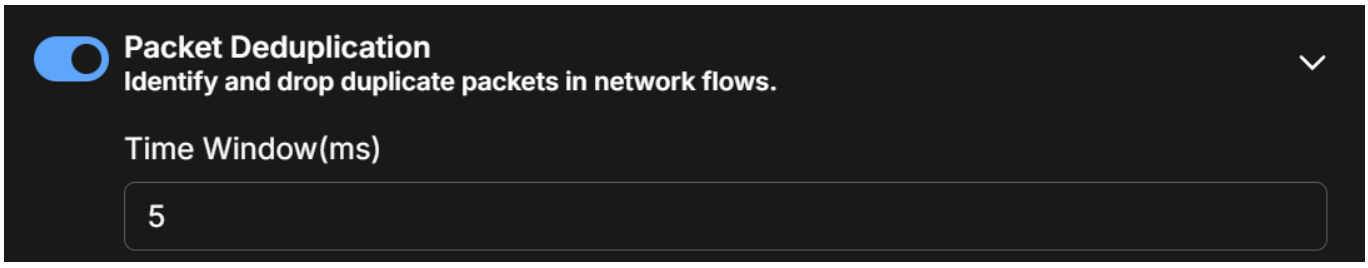


Enable this to build correlations between processes running on the sensor and host, and the IP address/port visible in traffic. The processes monitored include:

- log forwarder
- IDS
- maltrace
- aella_flow
- SSHD
- HTTP

You can set the sampling time interval.

Packet Deduplication

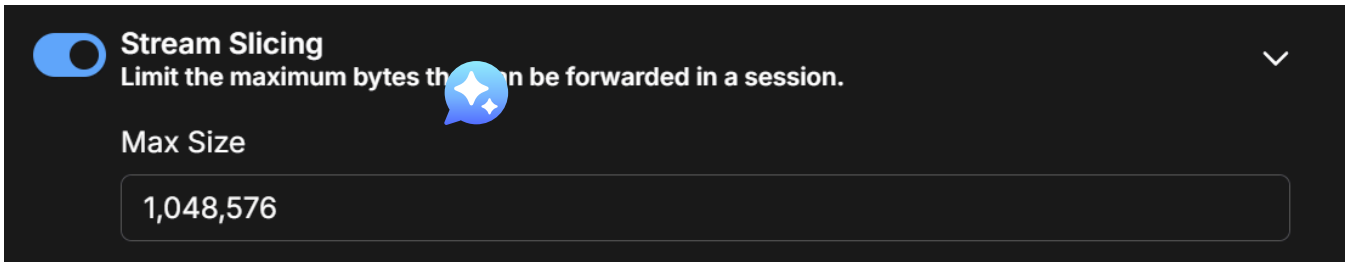


Enable this for the sensor to perform packet deduplication. This reduces storage by removing duplicate data. However, it uses processing power on the sensor to analyze the data, and can significantly slow system performance. When you enable it, you can set the deduplication time window.

Enable with caution.

If you have no need for this (for example, if you have a sensor for every network segment), disable it.

Stream Slicing



Enable this to perform stream slicing, which truncates sessions at the specified length. This can reduce the bandwidth used.

This has no effect if you use local file assembly.

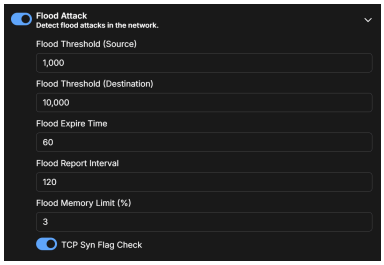
Handshake Failure



Enable this to detect handshake failures and allocate resources for that effort:

- **Time(s)** – Amount of time the sensor collects failures to get to the threshold
- **Threshold** – The number of failures over the specified time before an alert on this is triggered
- **Report Interval** – Interval between reports to the DP if there are multiple failures
- **Memory Limit (%)** – Maximum percentage of memory used by `aella_flow` for this detection

Flood Attack



Enable this to report flood attacks:

- **Flood Threshold (Source)** – The number of new session requests per second from a single source before an attack is reported
- **Flood Threshold (Destination)** – The number of new session requests per second from any source before an attack is reported
- **Flood Expire Time** – The number of seconds without new session requests after which we stop considering it an attack, and the sensor sends a `session_end` message to the DP
- **Flood Report Interval** – During an attack, the interval at which the sensor reports the attack to the DP (so that the DP isn't flooded with reports of the flood attack)

- **TCP Syn Flag Check** – Only counts session requests if the SYN flag in the packet is set

Sensor Profile Customizations in the Sensor Tab

The settings in the **Sensor** tab let you configure:

- Buffering

Buffering

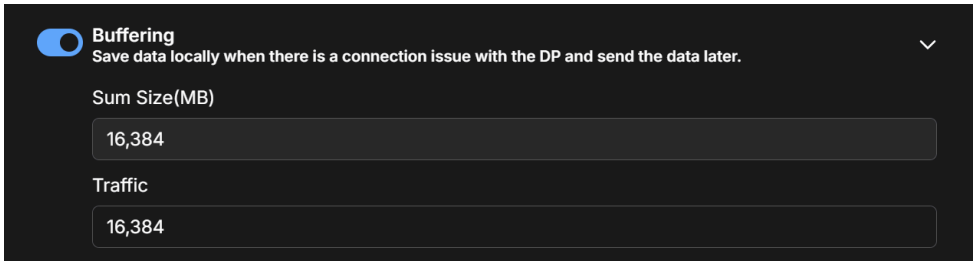
If you enable buffering, the sensor buffers the logs sent to Stellar Cyber. If the logs are received successfully, the sensor deletes the buffered logs. If the buffer is full, the sensor stops buffering logs.

> Windows events are buffered separately by winlogbeat and do not use the Buffering options. Similarly, FIM events are also buffered separately.

You can set the following buffering options:

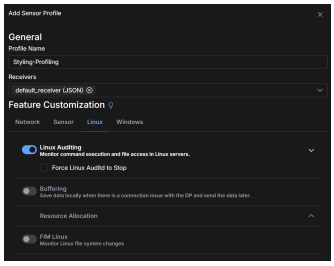
- **Sum Size (MB)** – The total of all **Traffic**, **Log Forwarder**, and **Maltrace** logs buffered
- **Traffic** – The amount, in MB, of traffic logs buffered

> When you disable buffering, any data in the buffer is immediately deleted. Ensure that the sensor has had a stable connection to the DP for long enough to transmit all data in the buffer.



Linux Server Sensor Customizations 6.1.0

The settings in the **Linux** tab let you specify settings for Linux Server Sensors:



- **Linux Auditing** – When enabled, the sensor checks its files for signs of alteration and monitors command executions.

  > This option is separate from the dedicated **FIM Linux** options for file integrity monitoring below.

- **Force Linux Auditd to Stop** – Stellar Cyber uses the `aella_audit` process to collect audit logs. Linux uses the `auditd` process. In most scenarios, the two processes coexist with each collecting data successfully. However, on older systems with pre-3.16 kernels, the processes can conflict, causing `aella_audit` to stop. If `aella_audit` is enabled, but you see no data on the DP, use this to stop the `auditd` process on older Linux systems.

  > In some situations, the `auditd` configuration may be locked by the Linux kernel (`audit configuration is locked in the kernel (enabled=2)`), preventing this option from working. In these situations, if you are running 4.3.7 sensor software, you must remove **immutable** flag (`-e 2`) from the `auditd` config, followed by a reboot of the host system. You **do not** need to do this in 5.1.1 – the **Force Linux Auditd to Stop** option works correctly, even if the `auditd` configuration is locked.

- **Buffering** – Enabling this allows you to set the size of the Traffic, Linux Event, and Windows Event buffers.

  > This option is deprecated but is still present to provide backwards compatibility with legacy Server Sensors that have it enabled in their profiles. It can no longer be enabled. However, legacy profiles that still have this feature enabled can use this option to disable it.

- **Resource Allocation** – You can set the following options:
  - **Control CPU Affinity** – Choose which CPU in a multi-core system the sensor uses for the control thread.
  - **Data CPU Affinity** – Choose which CPU in a multi-core system the sensor uses for the data processing thread.
  - **CPU Limit (%)** – Specify the amount of CPU per core that the sensor can use. To get the total for the sensor, multiply this percent by the number of cores. For example, if you set this at 5% and you have 8 cores, the sensor could use up to 40% of the CPU, as reported by the top and htop utility (distributed). If you have 32 cores, the sensor could use up to 160%. A total above 100% might seem impossible, but it's the sum of the usage percent on all the cores, each of which can have a maximum limit up to 100%.

    Note that the `aella_flow` process uses multiple threads. When using the `top` or htop utility with thread-mode enabled (`top H` or `htop H`), the output will show the percent that both the `aella_flow` process and its children threads use. At first, this might look like the CPU is processing twice as much as is configured because the individual threads add up to the same amount as their parent process and they're all shown. To see the output displayed hierarchically so you can visually distinguish data for the parent process from data for its children threads, enable tree view mode (`top V` or `htop F5`).

    The commands to use:

    - In `top`, press **H** to show threads and **V** to enable tree view.

    - In `htop`, press **H** to show threads and **F5** to enable tree view.

  By enabling thread display along with tree view, you can clearly see which threads are associated with which parent processes, providing a detailed and organized view of CPU usage.

  > The Server Sensor uses a minimum of 300 MHz, even if the value specified for **CPU Limit (%)** results in a smaller value based on the host CPUs. For example, if the Server Sensor is installed in a host with 2 CPUs running at 2.2 GHz and you specify a value of **5%** for **CPU Limit (%)**, the arithmetic of **5% * 2 CPUs * 2.2 GHz** would indicate an assignment of **220 MHz** for the Server Sensor. However, in a situation such as this, the Server Sensor uses its minimum assignment of 300 MHz, even though it is greater than the specified value.

  - **Memory Usage Limit (%)** – Specify the amount of memory the sensor can use.

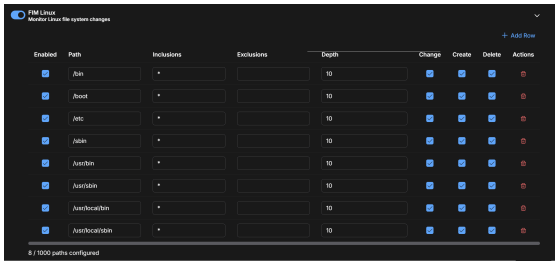
- **FIM Linux** – Enable this feature to monitor specified files and directories for changes, creations, and deletions. See the next section for details.

Configuring File Integrity Monitoring (FIM Linux)

> See the top of this page for recommended lessons about this section at Stellar Cyber Academy.

Enabling the **FIM Linux** feature in a Sensor Profile lets you keep track of changes to specified files and directories, including file changes, file creations, and file deletions.

When you enable the **FIM Linux** feature, a new table appears in the Sensor Profile dialog box that lets you specify both the files and directories you want to monitor for changes, as well as the types of changes for which you want to monitor. As you can see in the image below, the table is preconfigured with a set of paths recommended for monitoring.



Click the **+ Add Row** button to add new files/directories to the FIM feature. For each entry in the table, you can configure the following:

- **Enabled** – Specifies whether the indicated files/directories are enabled for monitoring. This lets you toggle a given file/directory on or off for monitoring without needing to delete it from the profile entirely.
- **Path** – Specifies the directory to monitor.
- **Inclusions** – Specifies the files within the directory to monitor. You can use the asterisk (*) character for standard wildcarding. For example, *.sh. If you leave this field blank, it

is implicitly set to *, meaning that all files in the specified path are included for monitoring.

- **Exclusions** – Specifies the files within the directory to exclude from monitoring. You may want to use this feature to exclude certain file types that are trusted and change often. Standard wildcarding is supported here, too. By default, this field is set empty and no files are excluded from monitoring.
- **Depth** – Specifies how many directories below the specified directory to monitor.

  A depth of 0 means that you are monitoring only the specified directory and nothing below it.

- **Change/Create/Delete** – Use these check boxes to specify the types of changes to track in the specified path.

In general, it's a good idea to optimize performance by focusing your FIM settings on specific files rather than large directories with many files and subdirectories. Refer to **Best Practices for File Integrity Monitoring (FIM)** for performance guidelines and a set of system files recommended for integrity monitoring.

The FIM feature works by recording a file hash for all files configured for monitoring in the Sensor Profile. Once the initial hash is recorded, the FIM feature regularly monitors for changes to the file hash and sends events to the DP if changes are seen.

You can keep track of events generated by the FIM feature in the following ways:

- Use the predefined Dashboards | PREDEFINED | File Integrity Monitoring dashboard.
- Use the Threat Hunting feature to search the Linux Events index for events with `msg_origin.source` set to `linux_agent` and `msg_origin.processor.type` set to `FIM` (or `agent.type` set to `auditbeat`). Once you find such an event, you can add it to the Threat Hunting Documents table by using the **Toggle add column to table** button adjacent to its entry and sort to see all such events.
- Similarly, you can create ATH rules with queries based on the `msg_origin.processor.type` field set to `FIM`.

The JSON for a FIM event indicates the type of change that took place. If the event indicates either a change or a deletion, the JSON includes both `old` and `current` values. For example, the illustration below shows a change in a file's hash between the `old` and `current` sections, indicating that its contents changed. If the event indicates a file creation, only the `current` section appears in the JSON.



Linux Server Sensor Installation Directories Excluded from FIM

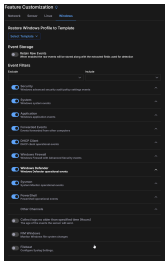
Note that the following installation directories for the Linux Server Sensor are automatically excluded from FIM monitoring. The user interface does not prevent you from adding these directories; however, they are still automatically excluded from FIM monitoring:

- `/var/aella/`
- `/var/log/aella`
- `/opt/aella`
- `/opt/aelladata`

Keep in mind that the FIM Linux feature can only monitor files and folders that exist when the `aella_audit` service restarts. If you add a non-existent file or folder for monitoring in the Sensor Profile and then create the folder later, it won't be monitored until `aella_audit` restarts.

Windows Server Sensor Customizations

See the top of this page for recommended lessons about this section at Stellar Cyber Academy.



You can configure Windows Server Sensors to collect specific logs. Logs are organized into different predefined channels in the **Windows** tab, each with its own maximum log size. You can toggle collection of the available logs and event types in each of the different individual channels, or opt for the simple route and use one of the predefined templates for Windows Server Sensors provided by Stellar Cyber. You can also apply filters configured in the System | DATA SOURCE MANAGEMENT | Data Filters | Log Filters page as Event Filters, excluding or including matching events.

Start with a Windows Server Sensor Template

Stellar Cyber recommends that you start configuration of a Windows Server Sensor profile by using one of the predefined templates included with the system. These templates have been carefully configured to match common deployment scenarios. Once you have reviewed the settings in a template and seen how it operates in your environment, you can tailor the settings in individual channels to fit your needs.

The following templates are available for options in the **Windows** tab of the ADD/EDIT SENSOR PROFILE window:

- **Windows Detect Profile (Low Volume)**. The selection covers the minimal events required for all native detections in Stellar Cyber.
- **Windows Context Profile (Medium Volume)**. Adds events commonly used by third-party detection rules.

- **Windows Compliance Profile (High Volume)**. Covers all Windows events.

Each of these profiles collects a different set of logs/events and results in a progressively higher volume of data ingestion from **Low** to **Medium** to **High**.

If you find that you are ingesting a higher volume of data than you would like relative to your license limits, you may want to reconfigure these settings, keeping in mind that the **Low Volume** profile provides enough coverage for all native Stellar Cyber detections.

Click here to see the differences in the logs/events collected by each of the available templates:

Channel			Notes	Windows Detect Profile	Windows Context Profile	Windows Compliance Profile
Security Collect Windows advanced security audit policy settings events.	**Account Logon Events**	Credential Validation	For UEBA alerts.	✔	✔	✔
		Kerberos Authentication Service	For UEBA alerts	✔	✔	✔
		Kerberos Service Ticket Operations		✔	✔	✔
		Other Account Logon Events				✔
	Account Management Events	Application Group				✔

Channel			Ms	Windows Detect Profile	Windows Context Profile	Windows Compliance Profile
		Management				
		Computer Account Management		✓	✓	✓
		Distribution Group Management				✓
		Security Group Management		✓	✓	✓
		User Account Management		✓	✓	✓
		Other Account Management Events		✓	✓	✓
	Detailed Tracking Events	DPAPI Activity		✓	✓	✓
		PNP Activity				✓
		Process Creation	For alerts related to proc	✓	✓	✓

Channel			Ns	Windows Detect Profile		Windows Context Profile		Windows Compliance Profile
			ess creation anomalies.					
		Process Termination				✓		✓
		RPC Events						✓
		Token Right Adjustment Events						✓
	DS Access Events	Detailed Directory Service Replication						✓
		Directory Service Access		✓		✓		✓
		Directory Service Changes			✓		✓	✓
		Directory Service						✓

Channel			Ms	Windows Detect Profile	Windows Context Profile	Windows Compliance Profile
	Logon/Logoff Events	Replication				
		Account Lockout				✓
		User/Device Claims				✓
		Group Membership				✓
		IPsec Extended Mode				✓
		IPsec Main Mode				✓
		IPSec Quick Mode				✓
		Logoff				✓
		Logon	For UEBA alerts	✓	✓	✓
		Network Policy Server				✓
		Special Logon			✓	✓
		Other Logon/L		✓	✓	✓

Channel			Ns	Windows Detect Profile	Windows Context Profile		Windows Compliance Profile
		ogoff Events					
	Object Access Events	Application Generated					✓
		Certification Services		✓	✓		✓
		Detailed File Share		✓		✓	✓
		File Share		✓	✓		✓
		File System		✓	✓		✓
		Filtering Platform Connection		✓		✓	✓
		Filtering Platform Packet Drop					✓
		Handle Manipulation					✓
		Kernel Object		✓	✓		✓
		Registry		✓	✓		✓
		Removable Storage					✓

Channel			Ms	Windows Detect Profile	Windows Context Profile	Windows Compliance Profile
		SAM			✓	✓
		Central Access Policy Staging				✓
		Other Object Access Events		✓	✓	✓
	Policy Change Events	Audit Policy Change		✓	✓	✓
		Authentication Policy Change		✓	✓	✓
		Authorization Policy Change		✓	✓	✓
		Filtering Platform Policy Change				✓
		MPSSVC Rule-Level Policy Change				✓
		Other Policy Change Events				✓

Channel			Ns	Windows Detect Profile	Windows Context Profile	Windows Compliance Profile
	Privilege Use Events	Non-Sensitive Privilege Use				✓
		Sensitive Privilege Use		✓	✓	✓
		Other Privilege Use Events				✓
	System Events	IPsec Driver				✓
		Security State Change		✓	✓	✓
		Security System Extension		✓	✓	✓
		System Integrity		✓	✓	✓
		Log Clear		✓	✓	✓
		Other System Events		✓	✓	✓
	Specify Event IDs	**Exclude** or		✓ **Include Only:** 1102, 1104, 4611, 4616, 4624-4625,	✓ **Include Only:** 1102, 1104, 4611, 4616, 4624-	

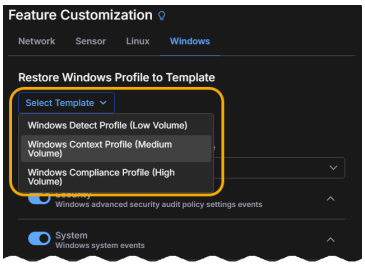
Channel			Ns	Windows Detect Profile	Windows Context Profile	Windows Compliance Profile
		Include Only		4648-4649, 4656-4658, 4660-4662, 4663, 4673-4674, 4688, 4768, 4692, 4697-4699, 4701-4702, 4704, 4706, 4713-4714, 4720, 4722, 4725-4732, 4734, 4738, 4741-4743, 4754, 4756-4758, 4765-4767, 4769, 4771, 4776, 4781-4782, 4794, 4799, 4825, 4898-4899, 4905, 5030, 5034-5035, 5038, 5136, 5140, 5145, 5156, 5379, 6281, 6416, 6423	4625, 4648-4649, 4656-4658, 4660-4663, 4672-4674, 4688-4689, 4692, 4697-4699, 4701-4702, 4704, 4706, 4713-4714, 4718-4720, 4722, 4724-4734, 4738-4743, 4754, 4756-4758, 4765-4767, 4768-4769, 4771, 4776, 4781-4782, 4794, 4799, 4825, 4898-4899, 4905, 5030, 5034-5035, 5038, 5136, 5140, 5145, 5156, 5379, 6281, 6416, 6423	
System Collect Windows system events.				✔	✔	✔
Application Collext Windows application events.				✔	✔	✔
Forwarded Events Collect Windows events forwarded from other Windows machines.				✔	✔	✔
DHCP Client Collect Windows DHCP client events				✔	✔	✔
Windows Firewall				✔	✔	✔

Channel			Ms	Windows Detect Profile	Windows Context Profile	Windows Compliance Profile
Collect Windows advanced security firewall events.						
Windows Defender Collect specified Windows Defender events.	**Specify Event IDs**			✓ **Include Only:** 1006-1008, 1015, 1116-1119, 2013	✓ **Include Only:** 1006-1008, 1013, 1015, 1116-1119, 1121, 2013, 5001, 5010, 5012-5013, 5101	✓
Sysmon Collect specified Windows sysmon events for process anomaly detections.	**Specify Event IDs**			✓ **Include Only:** 1, 3, 10, 13	✓ **Include Only:** 1, 3, 5, 7-8, 10-13, 15, 17, 20-21, 23	✓
PowerShell Collect specified Windows PowerShell operational logs.	**Specify Event IDs**			✓ **Include Only:** 4104	✓ **Include Only:** 4103-4104, 40961-40962	✓
FIM Configure and enable file integrity monitoring.	Not enabled by default in any template.					

How to Apply a Windows Server Sensor Template

You apply a profile for settings in the **Windows** tab as follows:

1. Navigate to **System | DATA SOURCE MANAGEMENT | Sensors | Sensor Profiles**, select **Create**, and then select **Add Standard Sensor Profile**.
2. Navigate to **Restore Windows Profile to Template** and choose the **Select Template** dropdown. Then, use the context menu that appears to select one of the available Windows Server Sensor profiles, as illustrated below:



Note the following:

- The **Set to Template** button is only available in the **Windows** tab of the Add/Edit Sensor Profile window and only affects the options available in the **Windows** tab.
- When you select one of the templates in the **Restore Windows Profile to Template** menu, all options in the **Windows** tab are reset according to the selected template. All other profile options are left unchanged. For example, if you have made customizations to the options in the **Network** tab, those options remain at their current settings after applying a Windows template to reset the options in the **Windows** tab.
- You can reset the options in the **Windows** tab using a template as often as you like, keeping in mind that all the options in the tab are reset to match the template each time you do so. You can also start with a template and then tweak individual settings as you see fit.

Applying Event Filters

You can use the **Add Event Filters** fields to **Exclude** or **Include** events matching the listed filters, giving you fine-grained control on which events are processed. You can apply any of the filters configured in the System | DATA SOURCE MANAGEMENT | Data Filters | Log Filters page as Event Filters; when you click or tap in the panels, a list of the available filters appears. Select any filters that you want to apply as **Exclude** or **Include** filters.

Keep in mind the following notes when using filters configured in the **System | DATA SOURCE MANAGEMENT | Data Filters | Log Filters** page as Event Filters for a Windows Sensor Profile:

- Event Filters apply to events in the following channels: **FIM (File Integrity Monitoring)** 5.4.0, **Security**, **System**, **Application**, **Forwarded Events**, **Microsoft Windows DHCP Client**, **Microsoft Windows Firewall with Advanced Security Firewall**, **Microsoft Windows Defender**, **Microsoft Windows Sysmon**, **Microsoft Windows PowerShell Operational**, and **Other Channels**.

- Event Filters are joined with the **Specify Event IDs** entries in the other channels using a logical `AND`. This means that a given event must match both the **Event Filters** and the **Specify Event IDs** entries to pass the filter.

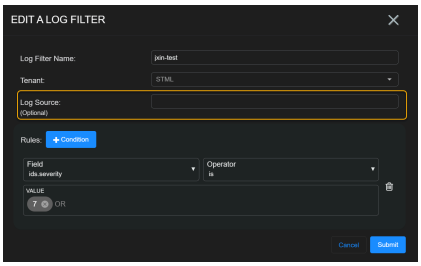
  For example, consider the following configuration:

  - **Event Filter** is set to `Exclude field process-id=100`
  - **Specify Event IDs** is set to `Include ID 1`

  In this situation the overall filter will ingest logs with an `ID of 1` that do not have `field process-id=100`.

  This allows you to apply advanced log filters to refine the data ingested by the Stellar Cyber platform, ensuring you focus on the most relevant events. For example, you can filter out event logs with particular process IDs detected by File Integrity Monitoring (FIM) while using a wildcard to match .exe filenames. If these processes are linked to routine system processes, excluding these logs prevents irrelevant data from consuming bandwidth and storage, while ensuring critical events from other processes are still ingested. This targeted filtering reduces unnecessary noise in your logs, helping you streamline investigations and focus on detecting meaningful threats.

- **Exclude** filters are always applied before **Include** filters, in the order they are specified.
- If you specify **Include** filters, only those logs matching the specified **Include** filters are ingested. All logs that do not match the specified **Include** filters are dropped.
- Only use Log Filters configured *without* the optional **Log Source** specified, as demonstrated in the figure below. Log Filters with a **Log Source** configured won't match any event logs on the Windows Server Sensor when used as Event Filters.

  Refer to **Managing Log Filters** for detailed instructions on configuring Log Filters.

- You can apply a maximum of 100 Event Filters each in the Exclude and Include panels.
- The Windows Server Sensor templates do not set any Event Filters.

Keeping Tabs on Log Filter Statistics

You can keep tabs on statistics related to log filters using the `show logfilter` and `show logfilter <filter-id>` commands in the sensor CLI. The `show logfilter-id` command shows the details of the last matching event during filtering and can be useful when evaluating filter performance.

In addition, log filter statistics are sent to the Sensor Monitoring index (`aella-ade-*`) on the DP using `msgtype:41` and can be viewed in the Threat Hunting page with **Indices** set to **Sensor Monitoring**. See below for the schema.

Expand to see the schema for log filter statistics

```
{
    "msgtype": 41,
    "msg_origin": {
        "processor": {
            "type": "log_forwarder", "winlog", "fim", "log_collector", or,
"aella_flow"
            "name": # Optional. It is <connector_name> when type is "connector".
            "id": # Optional. NG-SaaS, connector has id.
        }
    }
    "engid": # For connector running on DP, it is <connector type>_<connector
name>
    "engid_name": # For connector running on DP, it is <connector type>_<connector
name>
    "tenantid":
    "filter_id":
    "filter_name":
    "timestamp": # epoch ms
    "exclude_hits_delta": # Optional. The number of exclude hits of the filter by
the processor since last report
    "exclude_bytes_delta": # Optional. The size of excluded logs matched the
filter by the processor since last report
    "include_hits_delta": # Optional. The number of include hits of the filter by
the processor since last report
    "include_bytes_delta": # Optional. The size of included logs matched the
filter by the processor since last report
    "last_log": # Optional. The string of the latest log which was matched the
filter
}
```

Event Storage (Retain Raw Events) 6.5.0

You can enable the **Retain Raw Events** option if you want the sensor to store both raw and processed Windows event logs on the Stellar Cyber Platform (DP). This can be useful in troubleshooting situations when you may want to compare a processed event log to its corresponding raw log or when an investigation requires original event payloads.

You can search for raw Windows events by looking for the `message` field in the **Windows Events** index in the Interflow Search tab.

Keep in mind that when this feature is enabled, Windows Server Sensors store both raw and processed events, requiring more storage resources on the DP proportional to event volume.

About the Predefined Channels in the Windows Tab

The predefined channels allow you to collect events in the categories listed below:

- Refer to **Templates for Windows Server Sensors** for details on the different Windows Event IDs enabled in each of these categories for the available templates.
- Refer to **Alert Types That Use the Windows Index** for information on the alert types that use the events collected in these categories for alert generation.

Predefined Windows Channels:

- **Security** – Collect specified Windows advanced security audit policy settings events.
- **System** – Collect Windows system events.
- **Application** – Collect Windows application events.
- **Forwarded Events** – Collect Windows events forwarded from other Windows machines.
- **DHCP Client** – Collect Windows DHCP client events

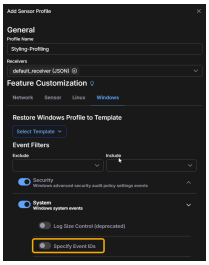
- **Windows Firewall** – Collect Windows advanced security firewall events.
- **Windows Defender** – Collect specified Windows Defender events.
- **Sysmon** – Collect specified Windows sysmon events for process anomaly detections.
- **PowerShell** – Collect specified Windows PowerShell operational logs.
- **Other Channels** – Collect additional specified Windows events that do not fit into the other categories.
- **FIM (File Integrity Monitoring)** – Perform FIM monitoring of specified files and directories.

All of the channels are enabled by default, with the exception of **FIM**. Events for each channel are stored in their own file. When a given file is full, older events are overwritten. If the sensor is receiving events faster than it can send them to the DP, you can increase the log file size to store more events or opt to exclude certain events.

Including or Excluding Specific Event IDs

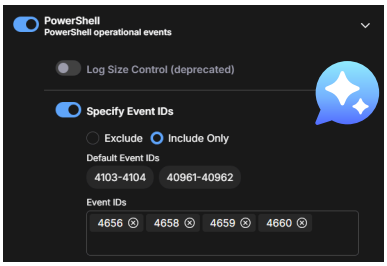
Each of the preset channels in the Windows tab includes a **Specify Event IDs** option that lets you either **Exclude** or **Include** specific Event IDs. (Note that the settings below these channels—**Collect logs no older than specified time (Hours)**, **FIM Windows**, and **Filebeat**—don't have this option.) Use the following procedure:

1. Navigate to the channel from which you want to include or exclude specific Event IDs and cascade its entry open.
2. Scroll to the bottom of the available options for the channel until you locate its **Specify Event IDs** entry. For example, the illustration below shows the **Specify Event IDs** entry for the System channel.



> Some of the channels have the **Specify Event IDs** option enabled automatically with IDs supplied for you if you have selected one of the available templates for the Windows tab options. Refer to **Templates for Windows Server Sensors** for a list of the options and Event IDs included in each of the available profiles.

3. Toggle the **Specify Event IDs** option on and choose whether you want to **Exclude** or **Include** specific IDs.
4. Supply the IDs you want to **Exclude** or **Include** by typing them into the **Event IDs** field, as illustrated below.



Note that the IDs you enter here are not validated. You can use the Event table in the Threat Hunting display (enable the column for Event IDs) to identify noisy/low value IDs, or refer to external resources, such as https://docs.microsoft.com/en-us/windows/security/threat-protection/windows-defender-application-control/event-id-explanations .

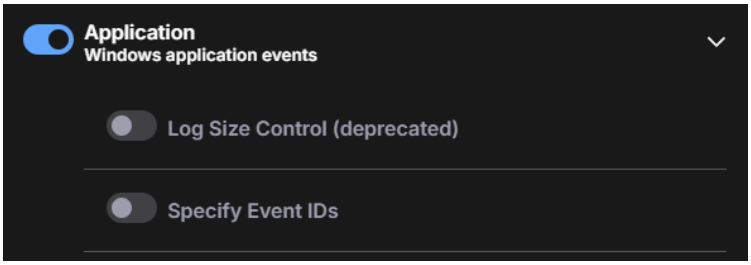
5. Click **Submit** when you are finished with all settings for the Sensor Profile.

Configuring the Log Size for a Channel (Deprecated)

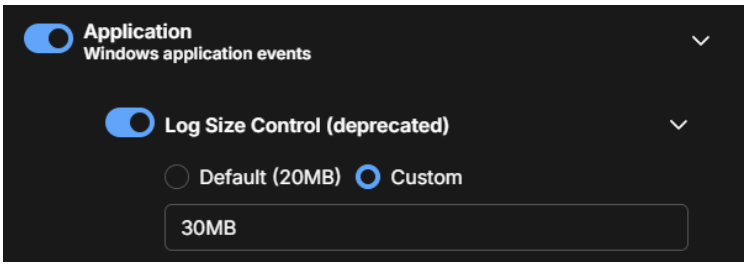
This feature was deprecated in the 5.5.0 release but could still be used. Starting with the 6.0.0 release, this feature is disabled on the Windows Server Sensor itself. Because of this, setting the **Log Size Control** option has no effect in sensor profiles applied to 6.0.0+ Windows Server Sensors.

To configure the log size for a channel:

1. Click on the name of the channel for which you want to customize log size.



2. Click on the **Log Size Control** toggle to enable it and to display size information.



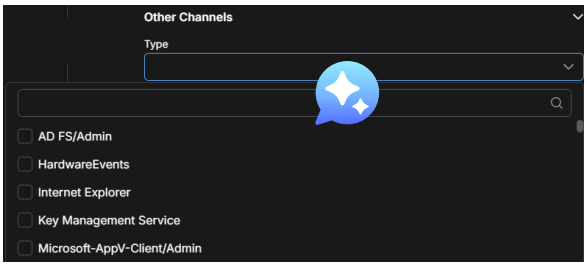
3. Click **Custom** to enable a custom log size.
4. Enter the custom size.
5. Click **Submit** after you finish all configurations.

Adding Other Channels

To add other channels:

1. Click **Other Channels**.

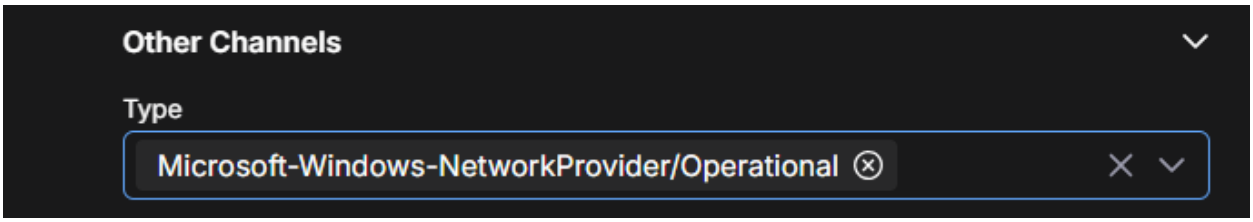
   The section expands.



You can either select from the available preset channels or type in your own.

2. To select from the preset channels, you choose one or more channels from the **Types** drop-down list and choose **Add**.

   The channel appears in the **Added** box.



   The available preset channels are extensive and are best reviewed in the user interface. In addition to many others, they include the following:

   - AD FS/Admin
   - Microsoft-Windows-AppLocker/EXE and DLL
   - Microsoft-Windows-AppLocker/MSI and Script
   - Microsoft-Windows-AppLocker/Packaged app-Deployment
   - Microsoft-Windows-AppLocker/Packaged app-Execution
   - Microsoft-Windows-GroupPolicy/Operational
   - Microsoft-Windows-Kernel-Boot/Operational
   - Microsoft-Windows-NetworkProvider/Operational
   - Microsoft-Windows-PowerShell/Admin
   - Microsoft-Windows-TCPIP/Operational
   - OpenSSH/Admin
   - OpenSSH/Operational
   - Windows Networking VPN Plugin Platform/Operational

3. You can also type in your own Windows event channel.

   The user interface checks to make sure the channel exists and reports an error message if it does not.

4. Select **Submit** after you finish all settings for the profile.

Limiting the Age of Collected Logs

To limit the age of the logs collected when you first enable:

1. Select **Collect logs no older than specified time (Hours)**.

The section expands.

2. Enter the number of h
3. Select **Submit** after you finish all settings for the profile.

Disabling All Windows Event Log Channels 6.2.0

If you disable all Windows event log channels in the sensor profile, the sensor's Winlogbeat service (**aella_winlog**) is automatically disabled, preventing unnecessary service execution. File Integrity Monitoring is unaffected and can still be run without enabling Winlogbeat.

The Winlogbeat service starts automatically once you configure at least one Windows event channel or a dependent feature requiring Windows logs is enabled. Note that disabling Winlogbeat stops Windows event-based detections, including Object Access events.

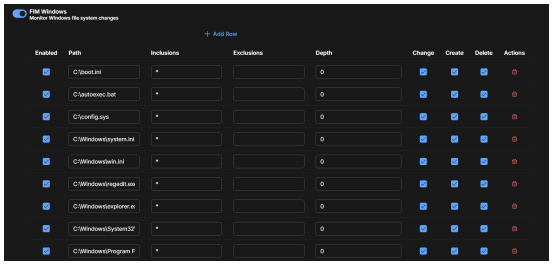
Configuring File Integrity Monitoring (FIM Windows)

> See the top of this page for recommended lessons about this section at Stellar Cyber Academy.

Enabling the **FIM Windows** feature in a Sensor Profile lets you keep track of file and folder creations, modifications, and deletions.

When you enable the **FIM Windows** feature, a new table appears in the Sensor Profile dialog box that lets you specify both the files and folders you want to monitor for changes, as well as the types of changes for which you want to monitor.

As you can see in the image below, the table is preconfigured with a set of files and folders recommended for monitoring by industry-standard best practices.



Select **+ Add Row** to add new files and folders to the FIM feature. For each entry in the table, you can configure the following:

- **Enabled** – Enable monitoring of the indicated files and folders. This lets you toggle monitoring for a given file or folder on or off without needing to delete it from the profile entirely.
- **Path** – Specify the full path to the folder that contains files and subfolders you want to monitor.

> A given path can only have one entry in the table. In addition, keep in mind that Windows paths are case-insensitive. Because of this, Stellar Cyber prevents you from adding, for example, both **C:\Program Files** and **C:\program files**. Because paths are not case-sensitive, Stellar Cyber treats these two paths as the same.

- **Inclusions** – List the files and subfolders within the specified folder to monitor. You can use an asterisk (*) for standard wildcarding. For example, *.bat, *.cfg, and so on. If you leave this field blank, it is implicitly set to *, meaning that all files and subfolders in the specified folder are included for monitoring.
- **Exclusions** – List the files and subfolders within the specified folder to exclude from monitoring. You might want to use this feature to exclude certain file types that are trusted and change often. Standard wildcarding is supported here, too. By default, this field is empty and no files are excluded from monitoring.
- **Depth** – Enter the number of subfolders below the specified folder to monitor.

  A depth of 0 means that you are monitoring only items in the specified folder and nothing below it.

- **Change/Create/Delete** – Select the types of activities to track in the specified folder.

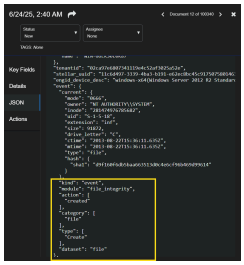
In general, it's a good idea to optimize performance by focusing your FIM settings on specific files rather than large folders with many files and subfolders (C:\, for example). Refer to **Best Practices for File Integrity Monitoring (FIM)** for performance guidelines and a set of system files recommended for integrity monitoring by Microsoft.

The FIM feature works by recording a file hash for each file selected for monitoring in the Sensor Profile. Once the initial hash is recorded, the FIM feature regularly monitors for changes to the file hash and sends events to the if changes are seen. For subfolders, FIM monitors changes in folder metadata to detect folder creation, deletion, and renaming.

You can keep track of events generated by the FIM Windows feature in the following ways:

- Use the predefined Dashboards | PREDEFINED | File Integrity Monitoring dashboard.
- Use the Threat Hunting feature to search the Windows Events index for events with `msg_origin.source` set to `windows_agent` and `msg_origin.processor.type` set to `FIM`. Once you find such an event, you can add it to the Threat Hunting Documents table by using the **Toggle add column to table** button adjacent to its entry and sort to see all such events.
- Similarly, you can create ATH rules with queries based on the `msg_origin.processor.type` field set to `FIM`.

The JSON for a FIM event indicates the type of change that took place. If the event indicates either a change or a deletion, the JSON includes both `old` and `current` values. If the event indicates a file creation, only the `current` section appears in the JSON. For example, the JSON below shows a file creation event.



Note that when the **FIM Windows** feature is enabled on a Windows Server Sensor, a new **windows agent sensor fi** process is spawned on the host machine and can be viewed with other running processes.

Windows Server Sensor Installation Folders Excluded from FIM

Note that the following installation folders for the Windows Server Sensor are automatically excluded from FIM monitoring. The user interface does not prevent you from adding these folders; however, they are still automatically excluded from FIM monitoring:

- `C:\ProgramData\StellarCyber\Windows Agent Sensor\config`
- `C:\ProgramData\StellarCyber\Windows Agent Sensor\log`
- `C:\ProgramData\StellarCyber\Windows Agent Sensor\run`

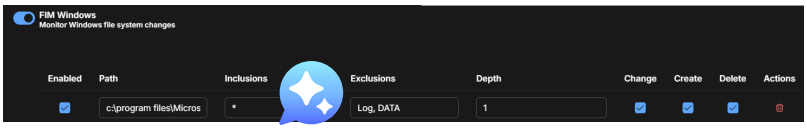
Keep in mind that the FIM Windows feature can only monitor files and folders that exist when the `windows agent sensor fim` service restarts. If you add a non-existent file or folder for monitoring in the Sensor Profile and then create the folder later, it won't be monitored until `windows agent sensor fim` restarts.

FIM Inclusion and Exclusion Example for a Windows Server Sensor Profiles

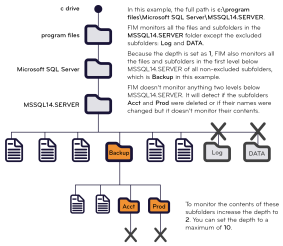
This example shows how to configure FIM to monitor the contents in a specified folder, exclude two subfolders, and monitor the contents of non-excluded subfolders.

1. Select **System | DATA SOURCE MANAGEMENT | Sensors | Sensor Profiles** and then either select **Create | Add Standard Sensor Profile** to create a new profile or select the **Edit** icon to edit an existing profile for a Windows Server Sensor.
2. If you're creating a new profile, enter a profile name, choose a tenant, choose **default_receiver**, and then select **Windows**.
3. Enable **FIM Windows** and then select **Add Row**.
4. In the newly added row, double-click or tap in the **Path** field and either type or paste the full path to the folder that you want to monitor, such as `c:\program files\Microsoft SQL Server\MSSQL14.SERVER`.
5. In the **Inclusions** field, leave the default setting (`*`)to monitor all files and subfolders in the **MSSQL14.SERVER** folder.
6. Double-click or tap in the **Exclusions** field and enter `Log, DATA` to exclude these two subfolders from monitoring.

   The **Exclusion** setting takes precedence over **Inclusion**.
7. Set Depth at `1` to monitor all files in the `MSSQL14.SERVER` folder and all files and subfolders in non-excluded first-level subfolders immediately below `MSSQL14.SERVER`.
8. Leave **Change**, **Create**, and **Delete** enabled to monitor for all these activities, leave the configuration as **Enabled**, and then **Submit**.



The monitoring behavior that these settings define is shown in the illustration below.



Configuring Filebeat Settings

You can configure a Windows Server Sensor to act as a syslog forwarder, using its built-in Filebeat service to receive syslog records from a specified source and forward them to a Modular Sensor for parsing, filtering, and eventual shipment to the DP.

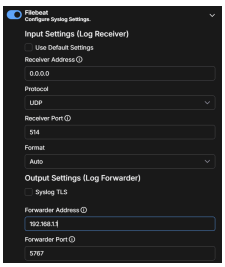
This feature lets you collect and ingest syslog records in situations where it might not make sense to deploy a dedicated Modular Sensor for syslog ingestion. The **Filebeat** feature is also supported on both Windows desktop and server versions, allowing you to forward logs from sites without server infrastructure.

> Using the Server Sensor as a lightweight log forwarder in a site without server infrastructure **is the only scenario** where the Windows Server Sensor is supported on a Windows desktop host. Deployments such as these support both Windows 10 and 11 desktop versions. Keep in mind, however, that when using this feature on a Server Sensor installed on a Windows desktop, you must enable **only** the **Filebeat** option in the **Windows** tab of the Standard Sensor Profile applied to the sensor.

Use the following procedure to configure a Windows Server Sensor as a log forwarder for syslog records:

1. Enable the **Filebeat** option.

   Once you enable **Filebeat**, new options appear that let you configure both **Input** and **Output**settings for the syslog forwarding feature.

2. The **Input Settings (Log Receiver)** options let you specify where the Windows Server Sensor listens for incoming syslog records. Set the following options:

   - **Receiver Address** – Specify the IP address where the Windows Server Sensor receives incoming logs.

     By default, this option is set to 0.0.0.0, which means that the Windows Server Sensor listens on all interfaces on the host Windows system. If your sensor is

installed on a Windows host with multiple interfaces and you only want to listen on one of them, you can specify which one using this field.

- **Protocol** – The transport protocol used for the syslog records. The default is UDP but you can also use TCP if your log source uses that protocol.
- **Receiver Port** – The port on which to receive syslog records. The default is the standard syslog port of 514.

  You must open this port in any firewalls or access control lists between the Windows Server Sensor and the sending log source.

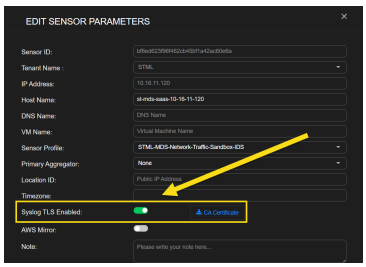
- **Format** – You can select format of the syslog records (RFC 3164 or RFC 5424), or you can leave this option at its default value of **Auto** so that the Sensor identifies the format automatically.

  You can check the **Use Default Settings** options if you want to return to the Stellar Cyber defaults for the **Input Settings**.

3. The **Output Settings (Log Forwarder)** options let you specify where the Windows Server Sensor forwards received syslog records. Set the following options:

   - **Syslog TLS** – Enable this option if you want to forward the logs with TLS encryption. Use the following procedure:

     a. Enable the **Syslog TLS Enabled** option in the Edit Sensor Parameters dialog box for the Modular Sensor to which you want to forward logs. Display this dialog box by selecting **Edit** for the Modular Sensor in the **System | DATA SOURCE MANAGEMENT | Sensors | Sensors** list.

     b. Enable the **Syslog TLS** option here in the **Filebeat** options in the **Windows** tab in the Standard Sensor profile.

     With these options configured, both the sending Windows Server Sensor and the receiving Modular Sensor download a self-signed certificated from the DP and use it for secure communications.

   - **Forwarder Address** – Specify the IP address of the Modular Sensor to which you want to forward the logs received by the Windows Server Sensor.
   - **Forwarder Port** – Specify the port on the receiving Modular Sensor to which the Windows Server Sensor should send the forwarded logs. You can leave this field blank if you would like the Windows Server Sensor to send the logs on the same port that the log source used.

4. Make sure the **Receiver Port** you specified for the **Filebeat** feature is open in the Windows Defender Firewall for the Windows Server Sensor host system (or in any other intervening firewalls or access control lists, depending on your implementation).
5. Configure your syslog source to send logs to the IP address and port specified in the **Input Settings** for the **Filebeat** feature.

Notes on Using the Windows Server Sensor as a Log Forwarder

Keep in mind the following important points when using a Windows Server Sensor to forward syslog records:

- The Windows Server Sensor performs minimal parsing on received records, only identifying the record format. Because of this, it cannot filter the syslog records it receives and simply forwards them all to the Modular Sensor. You must consider whether the convenience and cost savings of using a Windows Server Sensor instead of a Modular Sensor for log forwarding outweighs the potential network costs of sending all syslog records. The table below summarizes the pros and cons of each approach:

Sensor Type	Summary	Con
Windows Server Sensor	Inexpensive per-instance. Can be installed in Windows workstation versions; no server infrastructure required.	Cannot filter incoming records. All records forwarded to destination, potentially resulting in higher network costs.
Modular Sensor	Can filter received records and only ship what's necessary to the Stellar Cyber Platform.	More expensive than Windows Server Sensor. Requires dedicated VM or physical sensor.

- You must open the **Receiver Port** specified in the **Filebeat Input Settings** between the Windows Server Sensor and the log source in order for records to reach the sensor.
- When using the **Filebeat** feature on a Server Sensor installed on a Windows desktop, enable **only** the **Filebeat** option in the **Windows** tab in the Standard Sensor Profile. **All other options must be disabled**, as illustrated below:



- Use following procedure to use a Windows Server Sensor as a log forwarder for syslog records:
- You can use the `show logforwarder` command from the Sensor CLI to keep tabs on the configuration and performance of the feature.
- **Filebeat** and `dhcp_log` are mutually exclusive.

  The `dhcp_log` option is mutually exclusive with the Filebeat feature that lets Windows Server Sensors act as a forwarder for syslog records. You can use one or the other, but not both.

If you enable `dhcp_log` on a Server Sensor with the **Filebeat** option enabled, **Filebeat** is silently disabled and the `dhcp_log` setting is used.



Keep in mind that DHCP logs have been included with standard Windows Event Logs since the Windows Server 2012 release.

Memory Usage and Windows Server Sensor Profiles

When applying a sensor profile to a Windows Server Sensor, the system compares the amount of memory required by the profile to the amount of memory available on the target system and reports success or failure in the Sensors list as follows:

- **Systems with ≥ 100 GB Total Memory**: A Windows Server Sensor profile is applied successfully if the amount of free memory on the target system minus the memory required by the profile is ≥ 25 GB. If less, the failure is reported in the Sensors list.
- **Systems with < 100 GB Total Memory**: A Windows Server Sensor profile is applied successfully if the amount of free memory on the target system minus the memory required by the profile is ≥ 25% of the total memory. If less, the failure is reported in the Sensors list.

  If you receive a memory warning in the Sensors list, you can either reduce the memory footprint of the sensor profile (for example, by reducing the size of the Log Size Control options) or provision additional memory for the Windows Server Sensor host.

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